

Supply Chain X-Ray

Reference Data — Sample N-Tier Graph & Test Cases
SYN3RGY 3.0 · Open Innovation Track

SAMPLE GRAPH — TESLA INC. X EV TRACTION MOTOR (HSN 8501.53)

Company entered as Tier-0 input: Tesla Inc. (United States)

HSN selected from Tesla's import records: 8501.53 — AC multi-phase traction motors, output > 750W
Graph traces recursively from Tesla's direct suppliers back through 6 tiers to terminal raw inputs. Each tier resolved via US CBP / ImportYeti / UN Comtrade and national trade records. BOM filtering applied at every hop.

Tier 0 — Company Input

Tier	Company	Country	HSN Code	Commodity	Note
T0	Tesla Inc.	USA	8501.53	AC Traction Motors (multi-phase, >750W)	Anchor company — HSN selected from import history

Tier 1 — Direct Suppliers

Resolved from Tesla inbound shipment records. Motor-relevant suppliers retained after BOM filtering.

Tier	Company	Country	HSN Code	Commodity	Note
T1	Nidec Corporation	Japan	8501.53	EV Traction Motor Assemblies	Primary motor supplier; ships to Tesla Fremont & Gigafactory
T1	BorgWarner Inc.	USA	8501.53	eMotor Drive Units	Integrated motor-inverter units for drive systems
T1	Moog Inc.	USA	8503.00	Motor Stator & Rotor Components	Precision wound stator assemblies
T1	Denso Corporation	Japan	8501.53	Motor Assemblies (EV-grade)	Long-term Tesla partner; axle motor units
T1	Hitachi Astemo	Japan	8501.40	DC Motors & Actuator Assemblies	Formerly Hitachi Automotive; axle-integrated units
T1	Remy International	USA	8503.00	Stator & Rotor Sub-assemblies	Specialist EV stator winding; domestic US supplier

Tier 2 — Foreign Shippers / Sub-Suppliers

Resolved from Tier-1 import records. HS codes may shift at national sub-classification level when crossing borders.

Tier	Company	Country	HSN Code	Commodity	Note
T2	Nidec Dalian Co.	China	8503.00	Motor Stator Assemblies	Manufacturing arm of Nidec; ships wound stators

Tier	Company	Country	HSN Code	Commodity	Note
T2	Sumitomo Electric	Japan	7408.11	Copper Winding Wire (>6mm)	Magnet wire for motor coils; JIS C 3202 grade
T2	POSCO	South Korea	7225.19	Flat-rolled Silicon-Electrical Steel	Rotor & stator lamination stock; HGO grade
T2	Valeo SA	France	8503.00	Rotor Lamination Stacks	Precision stamped electrical steel laminations
T2	Proterial Ltd.	Japan	7408.19	Fine Copper Conductor Wire	Formerly Hitachi Metals; specialty winding wire
T2	Furukawa Electric	Japan	7408.11	Copper Magnet Wire	Enamelled copper wire for motor windings
T2	Nippon Steel Corp.	Japan	7225.19	Grain-oriented Electrical Steel	JNEX-Core grade; ultra-low loss for EV motors
T2	Thyssenkrupp Steel	Germany	7225.19	Non-oriented Electrical Steel (NOES)	Motor-grade flat-rolled; ships to EU fabricators
T2	LS Cable & System	South Korea	7408.11	Industrial Copper Wire Rod	Drawn copper rod for winding wire production
T2	Xinjiang Goldwind	China	8503.00	Permanent Magnet Motor Components	Rotor magnet assemblies; rare-earth NdFeB based

Tier 3 — Component Material Producers

Upstream of Tier-2 fabricators. Semi-processed and refined materials feeding motor winding and core steel production.

Tier	Company	Country	HSN Code	Commodity	Note
T3	Jiangxi Copper Co.	China	7403.11	Refined Copper Cathodes (99.99%)	Primary copper refiner; feeds wire drawing plants globally
T3	Aurubis AG	Germany	7407.10	Continuous Cast Copper Rod	Largest copper recycler in EU; rod for extrusion
T3	NLMK Group	Russia / EU	7225.19	Electrical Steel Coils (grain-oriented)	Cold-rolled silicon steel for core laminations
T3	Baotou Steel	China	7225.11	Grain-oriented Electrical Steel (GOES)	Transformer and motor grade silicon steel
T3	KME Group	Germany	7407.10	Copper Rod & Semifabricates	Downstream of smelters; specialty copper profiles
T3	Wieland Group	Germany	7408.19	Copper Strips & Fine Wire	Precision copper for EV winding applications

Tier	Company	Country	HSN Code	Commodity	Note
T3	ArcelorMittal	Luxembourg	7225.19	Electrical Steel Sheet (NOES)	Wide-strip NOES for motor laminations; global supply
T3	Cleveland-Cliffs	USA	7225.19	Grain-oriented Electrical Steel (AK)	Only US-based GOES producer; AK Steel subsidiary
T3	Ningbo Tongding	China	7408.11	Drawn Copper Wire (fine gauge)	High-volume magnet wire for motor winding
T3	Inner Mongolia Copper	China	7403.11	Refined Copper Cathodes	State-linked refiner; feeds Jiangxi & Nidec Dalian

Tier 4 — Raw Material Producers

Primary ore and metal producers. Trade records become sparser here — some nodes are last resolvable in open data.

Tier	Company	Country	HSN Code	Commodity	Note
T4	Codelco	Chile	2603.00	Copper Ore & Concentrates	World's largest copper producer; feeds Aurubis & Jiangxi
T4	BHP / Escondida	Chile	2603.00	Copper Ore & Concentrates	Escondida is single largest copper mine globally
T4	Freeport-McMoRan	USA / Indonesia	2603.00	Copper Ore & Concentrates	Grasberg mine; major global copper source
T4	Glencore	Switzerland / DRC	2603.00	Copper Ore & Concentrates	Katanga & Mutanda mines (DRC); cobalt by-product
T4	Antofagasta plc	Chile	2603.00	Copper Ore (Los Pelambres)	Mid-tier copper miner; feeds Asian smelters
T4	Anglo American	UK / Chile	2603.00	Copper Ore (Los Bronces)	Integrated miner; concentrate to Aurubis pipeline
T4	Zijin Mining Group	China	2603.00	Copper Ore & Gold Concentrate	Owns Timok & Buritica mines; growing global footprint
T4	Vale S.A.	Brazil	2601.11	Iron Ore (non-agglomerated)	Feeds POSCO, NLMK, ArcelorMittal blast furnaces
T4	Rio Tinto	Australia / UK	2601.11	Iron Ore (Pilbara fines & lump)	Largest iron ore operation; Western Australia

Tier	Company	Country	HSN Code	Commodity	Note
T4	Fortescue Metals	Australia	2601.11	Iron Ore (Pilbara)	Third-largest iron ore exporter; feeds East Asian mills

Tier 5 — Mining Inputs & Extraction Services

Suppliers of equipment, chemicals, and energy inputs consumed directly in mining and ore processing operations.

Tier	Company	Country	HSN Code	Commodity	Note
T5	Sandvik AB	Sweden	8430.31	Drilling & Rock-cutting Equipment	Tunnelling & open-pit drilling rigs for copper mines
T5	Caterpillar Inc.	USA	8429.52	Mining Trucks & Excavators	797F haul trucks; dominant at Escondida & Grasberg
T5	Komatsu Ltd.	Japan	8429.52	Ultra-class Mining Machinery	930E electric-drive haul trucks; competes with CAT
T5	Orica Ltd.	Australia	3602.00	Mining Explosives (ANFO / Emulsions)	Largest explosives supplier to open-cut copper mines
T5	BASF SE	Germany	3824.99	Flotation Chemicals & Reagents	Xanthate collectors; copper sulphide flotation process
T5	Cytec Solvay	Belgium	3824.99	Mineral Processing Reagents	Specialty frothers and depressants for copper circuits
T5	Nalco Water (Ecolab)	USA	3824.99	Process Water Treatment Chemicals	Scale/corrosion inhibitors for heap leach operations
T5	Atlas Copco	Sweden	8425.31	Compressors & Underground Equipment	Pneumatic & hydraulic tools for underground mining
T5	FLSmidth	Denmark	8474.20	Crushing & Grinding Mills	SAG & ball mills for ore comminution at copper plants
T5	Metso Outotec	Finland	8474.20	Mineral Processing Equipment	Flotation cells, thickeners, smelting furnace tech

Tier 6 — Raw Inputs to Mining Operations (Terminal Tier)

Furthest upstream resolvable tier. Commodity and industrial inputs consumed by extraction operations themselves.

Tier	Company	Country	HSN Code	Commodity	Note
T6	Air Products & Chemicals	USA	2804.40	Industrial Oxygen (for smelting)	Oxygen-enriched blast for copper smelter flash furnaces
T6	Linde plc	Ireland / Germany	2804.40	Industrial & Specialty Gases	Nitrogen, argon, oxygen supply to processing plants
T6	Shell / BP	Netherlands	2710.19	Diesel & Heavy Fuel Oil	Mine fleet fuel; bulk delivered to Atacama & Pilbara
T6	Dyno Nobel	Australia / USA	3602.00	ANFO & Emulsion Explosives (bulk)	Upstream of Orica; precursor ammonium nitrate supply
T6	Vulcan Materials	USA	2517.10	Grinding Media (steel balls/pebbles)	Consumable grinding media for ball mill circuits
T6	Saint-Gobain	France	6902.20	Refractory Bricks & Linings	Furnace lining for copper smelters and converters
T6	Yara International	Norway	3102.10	Ammonium Nitrate (ANFO precursor)	Fertiliser-grade AN; converted to mining explosive
T6	Siemens Energy	Germany	8504.40	Static Converters / Drive Systems	Variable frequency drives for SAG mills & conveyors
T6	ABB Ltd.	Switzerland	8501.64	High-voltage AC Motors (mill drives)	Gearless mill drive motors for grinding operations
T6	Hitachi Energy	Switzerland	8504.21	Power Transformers (mine substation)	HV transformers powering mine electrical infrastructure

HSN CODE CROSS-COUNTRY MAPPING

The 6-digit WCO standard is universal. Digits beyond 6 vary by country — system must resolve the national code when crossing borders.

Commodity	WCO 6-digit	US HTS	IN HSN	Notes
AC Traction Motors	8501.53	8501.53.4000	8501.5300	US appends 4 digits; India uses 8-digit
Copper Wire >6mm	7408.11	7408.11.6000	7408.1100	Consistent across US/IN at 6-digit level
Silicon-Elec. Steel	7225.19	7225.19.0090	7225.1900	US 10-digit HTS adds end-use suffix